

Lecture #22

Content to be covered

- ❖ de-Broglie wave
- ❖ Davission - Germer Experiment

Resources to be consulted

- ❑ Concepts of Modern Physics by Arthur Beiser

de-Broglie wave

- de-Broglie wave is also called “**matter wave**”.
- **Wave: Any aspect of the behavior or property of a material objects that varies with time and/or space in confirmation with the mathematical equations, describe a waves.**
- By analogy, with the wave and particle behavior of light that had already been established experimentally, the French physicist Louis de-Broglie suggested (1924) that particles might have wave properties in addition to particle properties.
- de-Broglie waves play an appreciable role, therefore, only in the realm of subatomic particles or light particles.

A moving body behaves in certain ways as though it has a wave nature

de-Broglie wave

- As we see that light comes in discrete units (Photons) with particles properties which is related to wave like properties having certain frequency and wavelength.
- In 1924, physicist Louis de-Broglie introduced the hypothesis that all material particle in motion possesses a wave character also. These waves are called “matter waves” with wavelength λ which are associated with the momentum p as below:

For a particle of momentum p and wave with frequency ν

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

$$\lambda = \frac{h}{\gamma mv} \quad \text{where} \quad \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

If E be the K.E. of the material particle, then

$$p = \sqrt{2mE} = \sqrt{2meV} = \sqrt{3mk_B T}$$

Characteristics of de-Broglie wave

- Lighter the particle, greater is the wavelength associated with it.
- Lesser the velocity of the particle, longer the wavelength associated with it.
- For $v = 0$, $\lambda = \infty$. This means that only with moving particle matter wave is associated.
- Whether the particle is charged or not, matter wave is associated with it. This reveals that these waves are not electromagnetic but a new kind of waves.

de-Broglie wave

Find the de-Broglie wavelengths of (a) a 46 gm golf ball with a velocity of 30 m/s, and (b) an electron with a velocity of 10^7 m/s.

(a) Since $v \ll c$, we can let $\gamma = 1$. Hence

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34} \text{ J} \cdot \text{s}}{(0.046 \text{ kg})(30 \text{ m/s})} = 4.8 \times 10^{-34} \text{ m}$$

(b) Again $v \ll c$, so with $m = 9.1 \times 10^{-31} \text{ kg}$, we have

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34} \text{ J} \cdot \text{s}}{(9.1 \times 10^{-31} \text{ kg})(10^7 \text{ m/s})} = 7.3 \times 10^{-11} \text{ m}$$

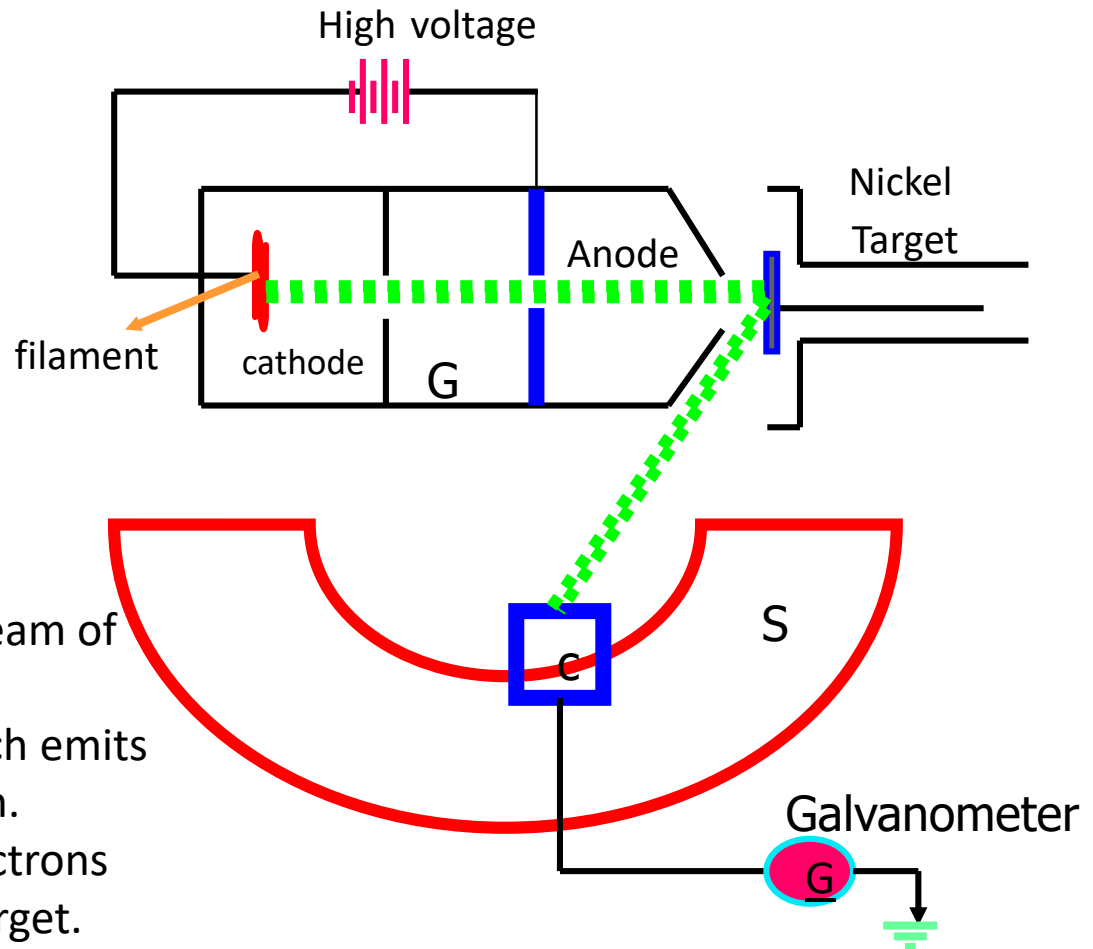
$$\lambda = \frac{h}{\gamma mv} = 7.304 \times 10^{-11} \text{ m}$$

Davission - Germer Experiment

Based on the concept of wave nature of matter fast moving electrons behave like waves.

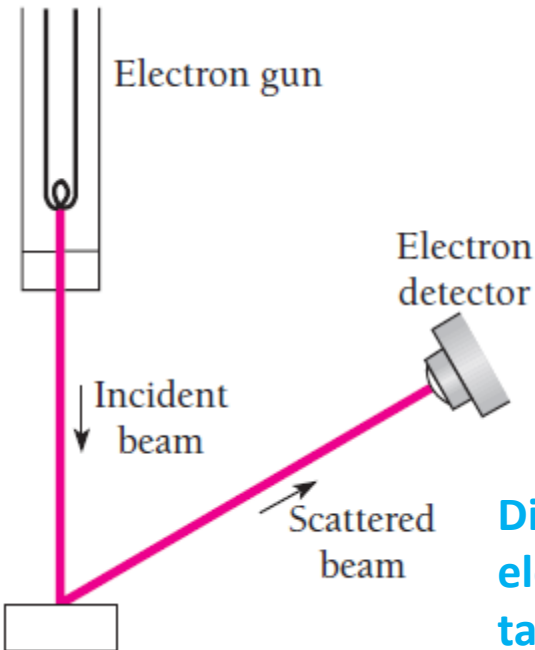
Hence accelerated electron beam can be used for diffraction studies in crystals

- The electron gun G produces a fine beam of electrons.
- It consists of a heated filament F, which emits electrons due to thermo ionic emission.
- The accelerated electron beam of electrons are incident on a nickel plate, called target.
- The target crystal can be rotated about an axis parallel to the direction of incident electron beam.



Davission - Germer Experiment

Experimental evidences for existence of de Broglie wave



Bragg's Eq. $2d \sin \theta = n\lambda$

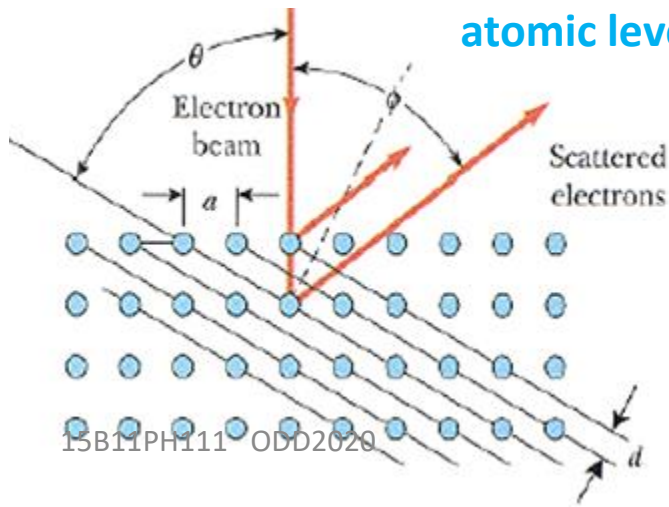
given $d = 0.91 \text{Å}$; $n = 1$ For wave $\lambda = 1.65 \text{Å}$

For matter wave $\lambda = \frac{h}{\sqrt{2meV}}$

For electron $\lambda = \frac{12.28}{\sqrt{V}} \text{Å}$ If $V = 54 \text{V}$: $\lambda = 1.67 \text{Å}$

Diffraction of electron beam takes place at atomic level

$$\theta = 90^\circ - \frac{\phi}{2}$$

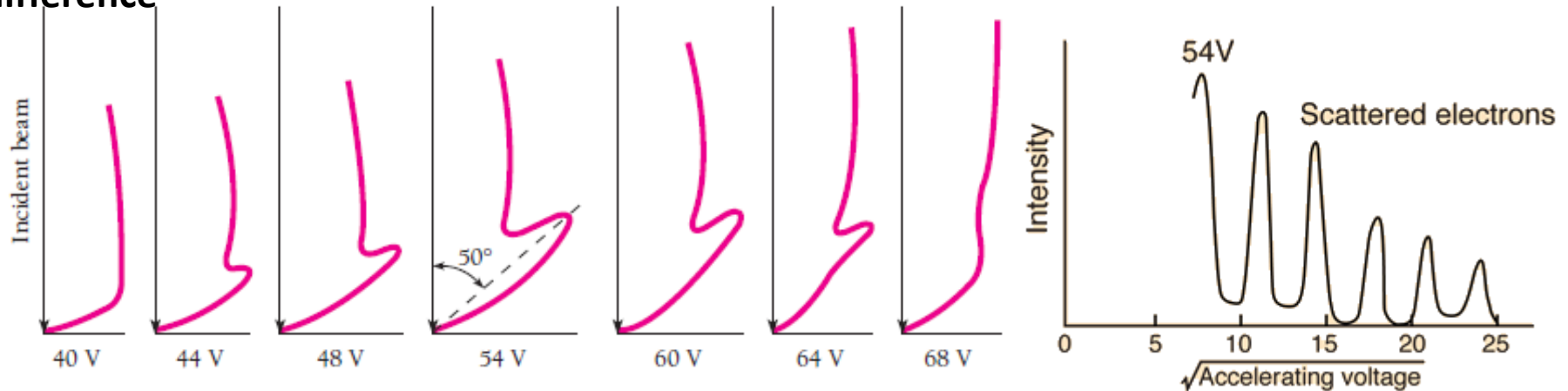


In 1927 Clinton Davisson and Lester Germer in USA and G. P. Thomson in England independently confirmed de Broglie's hypothesis **by demonstrating that electron beams are diffracted when they are scattered by the regular atomic arrays of crystals.**

Davission - Germer Experiment

Instead of a continuous variation of scattered electron intensity with angle, distinct maxima and minima were observed whose positions depended upon the electron energy!

Results of measurements of electrons entering into detector for various values of potential difference



There is a strong diffracting beam for $\Phi = 50^\circ$ and $V = 54$ Volts. $\lambda = 1.67 \text{ \AA}$

Davission - Germer Experiment

Now we use de-Broglie's formula $\lambda = h/\gamma mv$ to find the expected wavelength of the electrons. The electron kinetic energy of 54 eV is small compared with its rest energy m_0c^2 of 0.51 MeV, so we can take $\gamma = 1$.

So the electron momentum mv is

$$mv = \sqrt{2(9.1E-31)(54 \times 1.6E-19)} = 4 \times 10^{-24} \text{ kg.ms}^{-1}$$

Thus electron wavelength is given by $\lambda = \frac{h}{mv} = \frac{6.626E-34}{4.0E-24} = 0.166 \text{ nm}$

Thus, λ_{th} and λ_{exp} are nearly same and hence are in agreement which confirms de-Broglie's prediction.

The Davisson-Germer experiment thus directly verifies de-Broglie's hypothesis of the wave nature of moving bodies.

de-Broglie Wave Velocity

Wave or Phase velocity: Let us suppose, velocity of de Broglie's wave be v_p , then

$$v_p = v\lambda = \frac{E}{h} \cdot \frac{h}{p} = \frac{mc^2}{mv} = \frac{c^2}{v}$$

Since, the particle velocity 'v' is always less than 'c'; therefore, from above equation, it clear that “de Broglie's wave velocity” **v_p should always be greater than 'c' (the speed of light).**

It indicates that de Broglie wave associated with particle is leaving particle for behind. This is against the wave concept of particle and leading a discrepancy.